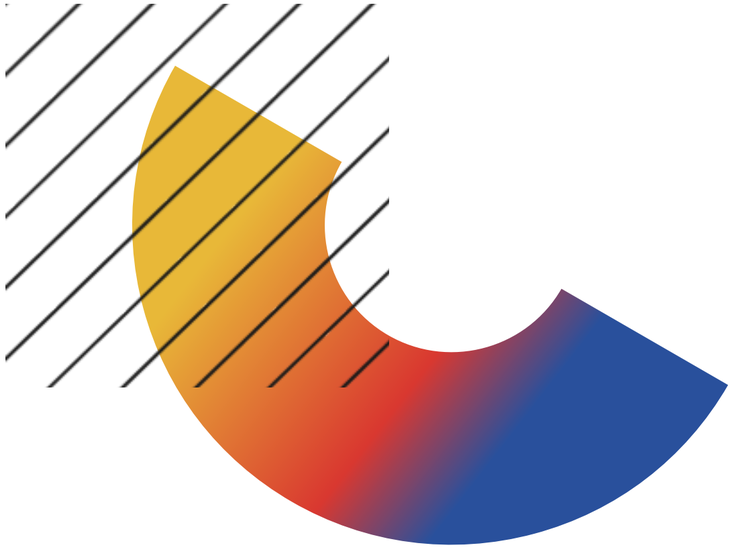
Maximizing Revenue

Agenda



- Problem Statement
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Problem Statement

In the fast-paced e-commerce sector, making the most of revenue is essential for long-term success.

Our goal is to use data-driven insights to maximise revenue streams. In order to meet this need our research aims to determine whether all the continents have consistent performace across all four quarters by focusing on the relationship regions, quarters and line sales.

Research Question

Is there a relationship between total fare amount and time of the day?

Can we nudge customers towards payment methods that generate higher revenue for drivers, without negatively impacting customer experience?

Data Overview

For this analysis, we utilized the comprehensive dataset of online_retail_II records, used data cleaning and feature engineering procedures to concentrate solely on the relevant columns essential for our investigation.

- Invoice
- StockCode
- Description
- Quantity
- InvoiceDate
- Price
- CustomerID
- Country

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
...	...	...	...	...	...	...	...	...
541905	581587	22899	CHILDREN'S APRON DOLLY GIRL	6	2011-12-09 12:50:00	2.10	12680.0	France
541906	581587	23254	CHILDRENS CUTLERY DOLLY GIRL	4	2011-12-09 12:50:00	4.15	12680.0	France
541907	581587	23255	CHILDRENS CUTLERY CIRCUS PARADE	4	2011-12-09 12:50:00	4.15	12680.0	France
541908	581587	22138	BAKING SET 9 PIECE RETROSPOT	3	2011-12-09 12:50:00	4.95	12680.0	France
541909	581587	POST	POSTAGE	1	2011-12-09 12:50:00	18.00	12680.0	France

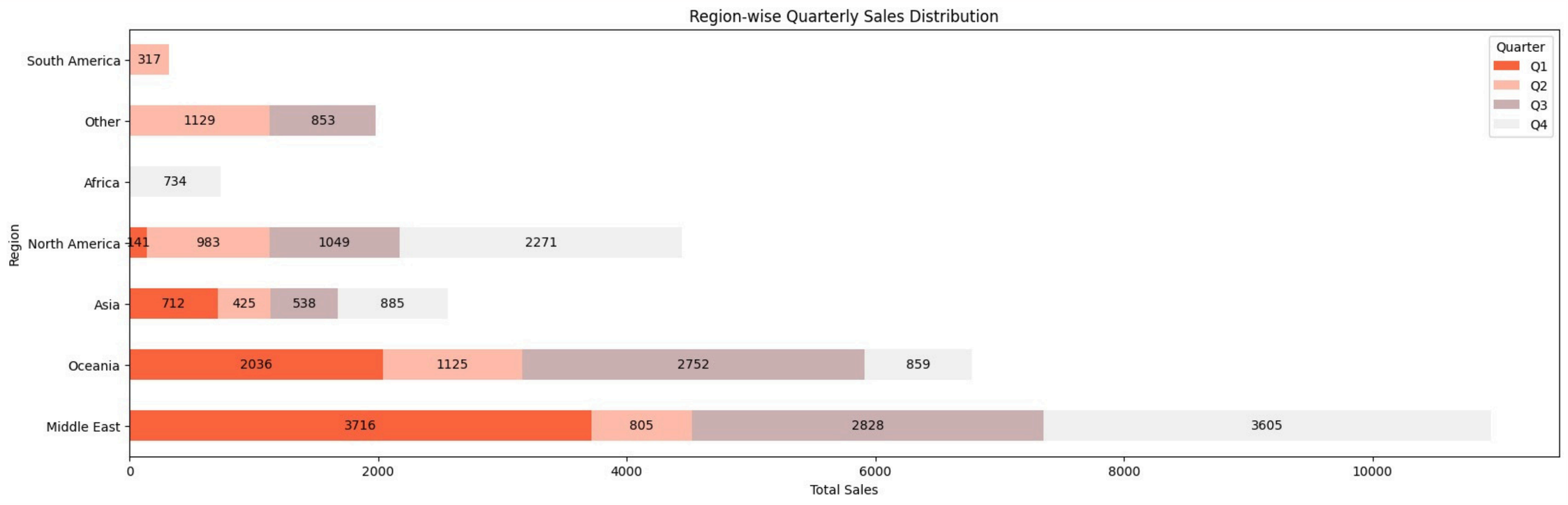
541910 rows × 8 columns

Methodology

Steps	Description
Descriptive Statistics	Performed statistical analysis to summarize key aspects of the data, focusing on fare amounts and payment types.
Hypothesis Testing	Conducted a ANOVA to evaluate the relationship between payment type and fare amount, testing the hypothesis that different regions and quarters influence line sale.

Insights

- Heavy dependence on Middle East and Oceania → potential concentration risk
- Opportunity to expand in underperforming regions (Africa, South America)
- Strong Q4 performance → optimize inventory and marketing strategies for peak season
- Region-specific strategies needed due to varying seasonal patterns



Hypothesis Testing

Null hypothesis: There is no siginificant difference between the aveage fares between customers from different continents for different quarters.
Alternative hypothesis: There is siginificant difference between the aveage fares between customers from different continents for different quarters.

With the p-value of 0.25, we fail to reject the null hypothesis and come to the conclusion that there is no significant difference between aveage fares between customers from different continents for different quarters.

Recommendations

- Encourage customers to make a successfull payment in the afternoon time to maximise revenues.
- Implement strategies such as offering incentives or discounts for credit card transactions to incentivize customers to choose their time of the day.

Thank You!!